

Solution 1.

$$1. a) \quad v_L = L \frac{di}{dt}$$

$$\text{so, } i = \frac{1}{L} \int_0^t v_L dt + i_L(0) \quad \text{--- (A)}$$

$$\text{Given } i_L(0) = 0 \text{ A}$$

$$\text{Also, it is given in the question } v_L(t) = \begin{cases} -3 & 0 \leq t < 1 \\ 6 & 1 \leq t < 2 \\ 0 & 2 \leq t < 3 \end{cases}$$

so, using eqⁿ (A),

$$\text{for } 0 \leq t < 1 \quad i = \frac{1}{3} \int_0^t -3 dt$$

$$i = -t + C_1$$

$$\text{as, } i(0) = 0 \text{ A}$$

$$\text{so, } \boxed{i(t) = -t}$$

for $1 \leq t < 2$

$$i = \frac{1}{3} \int 6 dt$$

$$i(t) = 2t + C_2$$

$$\text{As } i(1) = -1 \text{ A, so, } 2(1) + C_2 = -1 \Rightarrow C_2 = -3$$

$$\text{so, } \boxed{i(t) = 2t - 3}$$

for $2 \leq t < 3$

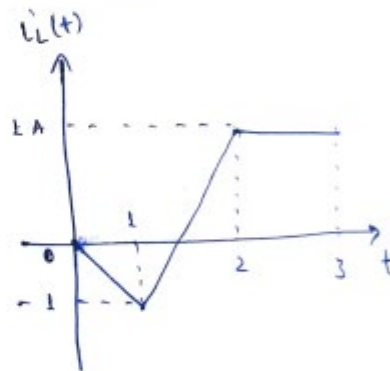
$$\dot{i} = \frac{1}{3} \int 0 \, dt$$

$$\dot{i} = 0 + C_3$$

as $i(2) = 1 \text{ A}$

so, $\boxed{\dot{i}(t) = 1 \text{ A}}$

so, current waveform will look like



b) $i = C \frac{dv_C}{dt} \Rightarrow v_C(t) = \frac{1}{C} \int_0^t i \, dt + v_C(0) \quad \text{--- (B)}$

from the current waveform as shown above, we can write

$$i_L(t) = \begin{cases} -t & 0 \leq t < 1 \\ 2t-3 & 1 \leq t < 2 \\ 1 & 2 \leq t < 3 \end{cases}$$

so, using eqn (B)

for $0 \leq t < 1$; $v_c(t) = 2 \int_0^t -t dt = -t^2 + c_1$

$$v_c(0) = 0 \Rightarrow c_1 = 0$$

so, $v_c(t) = -t^2$

for $1 \leq t < 2$; $v_c(t) = 2 \int_1^t (2t-3) dt = 2t^2 - 6t + c_2$

$$v_c(1) = -1 \Rightarrow c_2 = 3$$

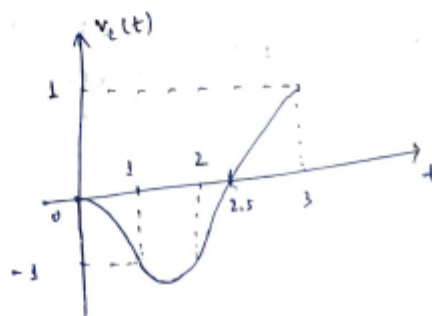
so, $v_c(t) = 2t^2 - 6t + 3$

for $2 \leq t < 3$; $v_c(t) = 2 \int_2^t 1 dt = 2t + c_3$

$$v_c(2) = -1 \Rightarrow c_3 = -5$$

so, $v_c(t) = 2t - 5$

so, the voltage waveform will look like



Solution 2)

We can simply apply the tricky concept here what we generally use with battery. So, if a current is leaving with positive terminal or entering with negative terminal, we simply say that power is supplied by the device means P_{out} . Just opposite if current is entering inside the device through positive terminal or leaving through negative terminal then we can say we are supplying power to the device means P_{in} .

Now coming to the question:

$$\overline{R_2}$$

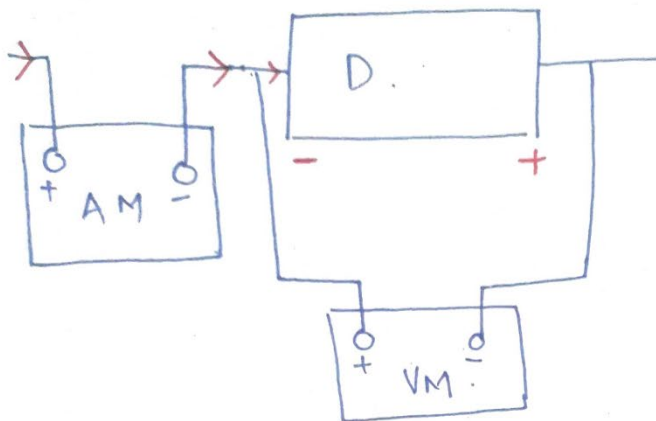


Figure a.

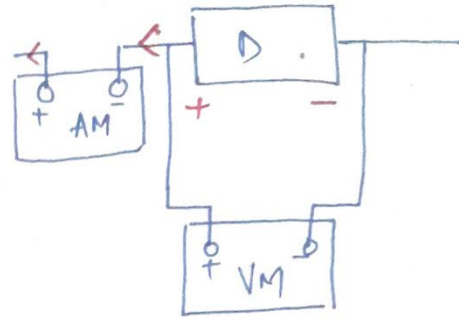


Figure b.

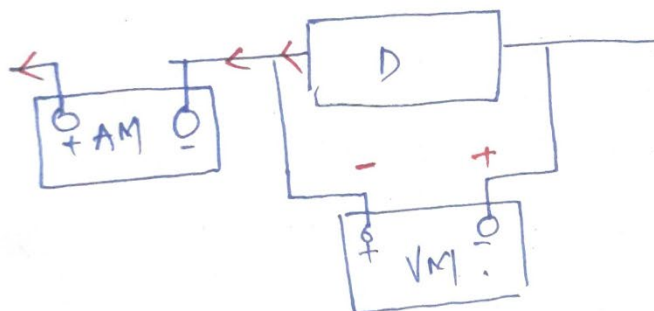


Figure c.

a) From the above figure(a), we can say that current is entering into the device through negative terminal of the device so we can say that power is flowing out of the device, or the power is supplied by the device.

b) From the above figure(b), we can say that current is leaving through the positive terminal of the device so we can say that power is flowing out of the device, or the power is supplied by the device.

c) From the above figure(c), we can say that current is leaving through the negative terminal of the device so we can say that power is flowing into the device, or the power is supplied to the device.

In other words,

a) OUT, since positive charges gain potential in flowing through D.

b) OUT, since positive charges gain potential in flowing through D.

c) In, since positive charges lose potential in flowing through D.

Solution 3)

a) Voltage source: $v(t) = 200t$ (for $t=0.5$ seconds)

Resistor: $R=200$

Capacitor: $C=1000 \mu\text{F}$

Time: $t=0.5$ seconds

$$\text{i) } P_R = iv = \frac{v^2}{R} = \frac{100^2}{200} = 50 \text{ W}$$

$$\text{ii) } P_C = iv = C \frac{dv}{dt}$$

$$v(t) = 200t$$

$$\frac{dv}{dt} = 200$$

$$\text{So, } i = 1000 * 10^{-6} * 200 = 0.2$$

$$P_C = 100 * 0.2 = 20 \text{ W}$$

iii) The total power delivered by the source is the sum of the power dissipated in the resistor and the power associated with the capacitor.

$$P_{\text{source}} = P_R + P_C = 50 + 20 = 70 \text{ W}$$

iv) The energy stored in a capacitor is given by:

$$W_C = \frac{1}{2} * C * v(t)^2$$

$$\text{At } t=0.5 \text{ seconds: } v(0.5) = 100$$

$$W_C = 5 \text{ J}$$

(i) Power dissipated in the 200Ω resistor: 50 W

(ii) Power dissipated in the $1000 \mu\text{F}$ capacitor: 20 W

(iii) Power delivered by the source: 70 W

(iv) Total energy stored in the capacitor: 5 J

b) Voltage source: $v(t)=2000t$ V

Time: $t=5$ ms = 0.005 s

Resistor: $R=4\ \Omega$

Capacitor: $C=300\ \mu\text{F}$

Current through a resistor is given by Ohm's Law:

$$i_R(t) = \frac{v(t)}{R}$$

At $t=0.005$ seconds:

$$v(0.005) = 2000 * 0.005 = 10\text{ V}$$

So,

$$i_R(0.005) = \frac{10}{4} = 2.5\text{ A}$$

Current through the 300 μF capacitor:

The current through a capacitor is given by:

$$i_c(t) = C \frac{dv(t)}{dt}$$

Given: $v(t) = 2000t$

$$\frac{dv(t)}{dt} = 2000$$

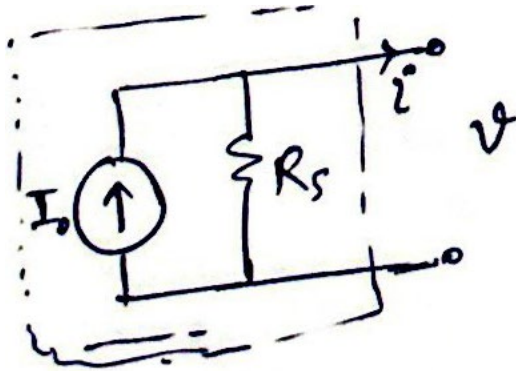
So,

$$i_c(t) = C * 2000 = 0.6\text{ A}$$

- Current through the 4Ω resistor at $t=5$ ms: 2.5 A
- Current through the 300- μF capacitor at $t=5$ ms: 0.6 A

Solution 4)

The solar cell needs to be modelled as current source as asked in the question. We know that an ideal current source will have a constant current for all voltage values. However, the $v - i$ characteristics given in the question is not of constant current. The $v-i$ characteristics up to 40mA given in the questions can be approximated as a straight line. Therefore, the current source we want to model would be non-ideal i.e. current source with a resistance in parallel as following:



It's $v - i$ relation can be written as: $i = I_0 - v/R_s$

When $i=0$, $v=1.2$

$$\Rightarrow 1.2 = I_0 \times R_s$$

$$R = \frac{1.2}{I_0}$$

When $i=40\text{mA}$, $v=1$

$$\Rightarrow 1 = (I_0 - 0.04) \times R_s$$

$$\Rightarrow 1 = (I_0 - 0.04) \times \frac{1.2}{I_0}$$

$$\Rightarrow I_0 = 0.24\text{A or } 240\text{mA}$$

$$R_s = \frac{1.2}{0.24}$$

$$R_s = 5 \text{ ohm}$$

Solution 5)

(a) $v = kq$

Here, v is the voltage and q is the charge.

Capacitance (C):

The relationship between voltage and charge in a capacitor is given by:

$$v = \frac{q}{C}$$

If we compare this with $v = kq$, we can identify that $k = \frac{1}{C}$.

$$\frac{dv}{dt} = k \frac{dq}{dt} = ki;$$

$$i = \frac{dq}{dt}$$

Capacitance with $C = \frac{1}{k}$

(b) $v = k \frac{dq}{dt} = ki$

Here, v is the voltage and $\frac{dq}{dt}$ represents the current i.

Ohm's Law states that $v = Ri$,

If we compare this with $v = k \frac{dq}{dt} = ki$, we can identify that $k = R$.

Therefore, this equation describes a resistor.

(c) $v = k \frac{d^2q}{dt^2} = k \frac{di}{dt}$

Here, v is the voltage and $\frac{d^2q}{dt^2}$ represents the rate of change of current $\frac{di}{dt}$

The relationship for an inductor is given by $v = L \frac{di}{dt}$

We can identify that $k = L$.

Therefore, this equation describes an inductor where $k = L$.

Solution 6)

We know that,

$$i = I = \frac{V_{avr}}{R},$$

$$W_R = \int_0^T Ri^2 dt$$

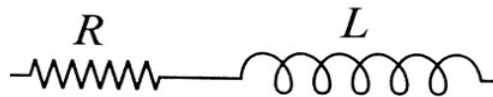
$$W_R = RI^2T = R \frac{V_{avr}^2}{R^2} \frac{1}{4} = \frac{V_{avr}^2}{Rf}$$

Since

$$W_c = \frac{1}{2} CV_{avr}^2 = 10W_R, \text{ or } C = \frac{20}{fR} = \frac{20}{60 * 10^4} = 33.3 \mu F$$

Solution 7)

Here we know that the coil can be best modeled by an inductor. Questions also says that it draws power in steady-state current. An ideal inductor should not absorb any power in steady current (since when $di/dt=0$, no voltage produced across L and therefore $v \cdot i=0$). Then it must be a non-ideal inductor which is an inductor with a series resistance as following.



$$V = L \frac{di}{dt} = L \frac{\Delta i}{\Delta t}$$
$$L = \frac{v}{\Delta i / \Delta t} = \frac{2 * 50 * 10^{-6}}{20 * 10^{-3}} = 5 \text{ mH}$$

Now the power is given by,

$$P = i^2 R \text{ or } R = \frac{P}{i^2} = \frac{0.2}{(10 * 10^{-3})^2} = 2 \text{ K}\Omega$$

Solution 8)

To derive the expression for the resistance R of the fuse in terms of the current I and evaluate the current at which the fuse "blows," follow these steps:

Given Relations:

$$\text{Resistance of the fuse: } R = R_R(1 + cT)$$

$$\text{Temperature rise: } T = kP, \text{ where } k \text{ is a constant, and } P \text{ is the power dissipated.}$$

$$\text{Power dissipated: } P = I^2R$$

Substitute Power into Temperature Rise:

$$\text{Since } P = I^2R, \text{ we have } T = kI^2R.$$

Substitute Temperature Rise into Resistance:

$$R = R_R(1 + c(kI^2R))$$

This simplifies to:

$$R = \frac{R_R}{1 - R_RckI^2R}$$

Current at Which the Fuse Blows:

The fuse "blows" when the resistance becomes very large (ideally infinite) so that no current flows through it (act as an open circuit).

This happens when the denominator of the above expression approaches zero:

$$1 - R_RckI^2R = 0$$

Solve for I :

$$R_RckI^2R = 1$$

$$I = \frac{1}{\sqrt{R_Rck}}$$

This is the critical current at which the fuse will melt and interrupt the circuit.

Solⁿ 9)

Battery capacity : 90 mA for 12 hr

$$\text{charge (Q)} = \int_0^t i dt = \int_0^{12 \text{ hr}} 90 \times 10^{-3} dt$$

$$= 90 \times 10^{-3} (12 \times 60 \times 60) = 3888 \text{ C}$$

$$V_t = 1.5 \text{ V} \Rightarrow \text{Energy} = \int_0^{12 \text{ hr}} V i dt$$

$$= 1.5 \times 90 \times 10^{-3} \times (12 \times 60 \times 60)$$

$$= 5.832 \text{ KJ}$$

$$\text{or in Wh} = \frac{5.832}{3600} = 1.6 \text{ Wh}$$

Solⁿ 10)

Charge is integration of current i.e.
area under the curve

$$\text{a) } t = 1 \text{ sec, } \Rightarrow \text{ total area} = 10 \times 1 = 10 \text{ C}$$

$$\text{b) } t = 3 \text{ sec, } \Rightarrow \text{ total area} = (10 \times 1) + \left(\frac{1}{2} \times 15 \times 1\right) + (5 \times 1) \\ = 22.5 \text{ C}$$

$$\text{c) } t = 5 \text{ sec, } \Rightarrow \text{ total area} = 22.5 + (5 \times 1) + \left(\frac{1}{2} \times 5 \times 1\right) \\ = 30 \text{ C}$$